A

Mini Project Report

on

Smart Parking Management System for APSIT

Submitted in partial fulfillment of the requirements for the

degree

Second Year Engineering – Computer Science Engineering (Data Science)

by

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UNIVERSITY OF MUMBAI

Academic year: 2025-26

CERTIFICATE

This to certify that the Mini Project report on **Smart Parking Management System for APSIT** has been submitted by **Swati Vishwakarma(24107035), Shivansh Shukla(24107019), Yash Shinde(24107076) and Shreyas Mhaske(24107017)** who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Computer Science Engineering (Data Science)**, during the academic year **2025-2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

Prof Harsha Zhope
Guide

Dr. Pravin Adivarekar
HOD, CSE(Data Science)

Dr. Uttam D. Kolekar
Principal

External Examiner:
1.

Internal Examiner:
1.

Place: A. P. Shah Institute of Technology, Thane
Date:

ACKNOWLEDGEMENT

This project would not have come to fruition without the invaluable help of our guide **Prof Harsha Zhope.** Expressing gratitude towards our HoD, **Dr. Pravin Adivarekar**, and the Department of Computer Science Engineering (Data Science) for providing us with the opportunity as well as the support required to pursue this project. We would also like to thank our project coordinator **Prof. Richa Singh and Prof Kiran Gaikwad** who gave us her valuable suggestions and ideas when we were in need of them. We would also like to thank our peers for their helpful suggestions.

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Chapter 1

Introduction

The increasing number of vehicles inside the APSIT campus has led to significant challenges in managing parking efficiently. Students and staff often face difficulty in finding available parking spaces, which results in time wastage, congestion, and improper parking practices.

1.1 Purpose

- To develop a smart parking management system for APSIT campus
- To provide real-time information about available parking slots
- To reduce time spent in searching for parking spaces
- To enable easy and quick reservation of parking slots
- To improve parking discipline among students and staff
- To ensure efficient utilization of parking space
- To minimize congestion and unorganized parking within the campus
- To enhance overall user convenience through a mobile application

1.2 Problem Statement

The APSIT campus currently faces challenges in managing parking efficiently due to the absence of a proper system. Users often struggle to find available parking spaces, leading to time wastage, congestion, and vehicles being parked in unauthorized areas. The lack of real-time information and monitoring results in unorganized parking and inefficient use of available space, highlighting the need for a smart and automated parking solution.

1.3 Objectives

- To design and implement a smart parking system using digital technologies

- To optimize parking space utilization through slot-based allocation
- To improve parking discipline using restricted zones
- To provide real-time parking availability and monitoring
- To enhance user convenience through a mobile application

1.4 Scope

- Applicable within APSIT campus for students and staff
- Enables quick parking slot discovery and reservation
- Can be expanded to other institutions or organizations
- Supports real-time monitoring and security improvements
- Reduces time spent searching for parking

Chapter 2

Proposed System

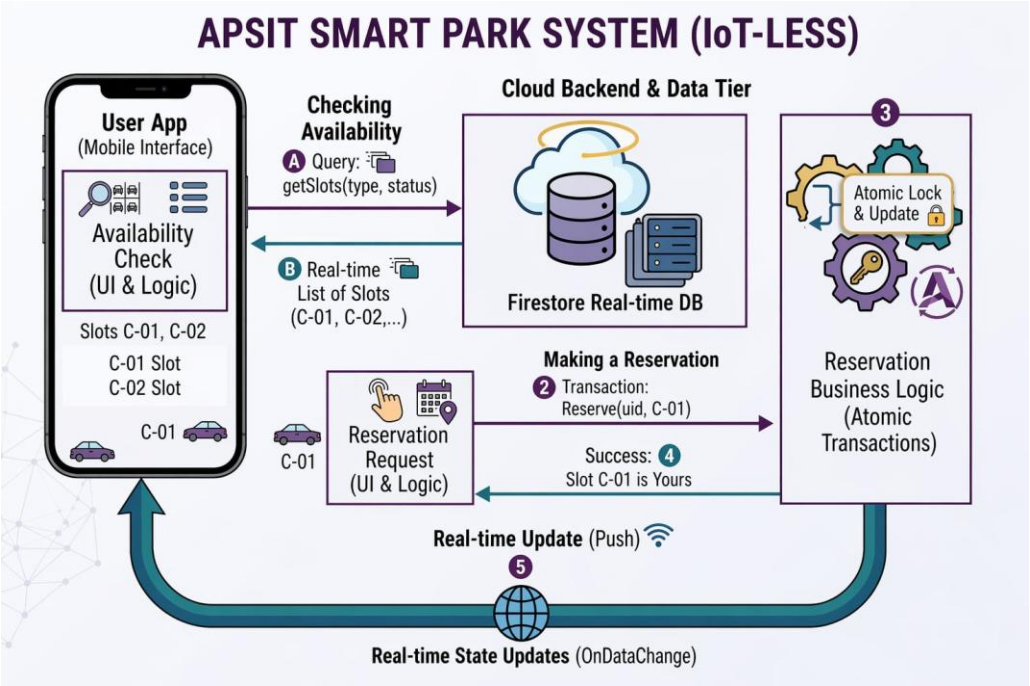


Fig 2 Block Diagram

Fig 2 indicates the Smart Parking Management System designed as a mobile application to streamline and automate parking operations within the APSIT campus. The system enables users to check real-time parking slot availability and reserve slots efficiently using a cloud-based backend.

The workflow begins when the user opens the app and checks parking availability through an interactive interface. The application sends a query to the backend (Firebase Firestore), requesting slot data based on parameters such as vehicle type and slot status. The backend responds with a real-time list of available parking slots, which is instantly displayed on the user's screen through a digital parking map.

Once the user selects a slot, a reservation request is sent to the backend. The system processes this request using atomic transactions to ensure that only one user can reserve a slot at a time, preventing double booking. If the reservation is successful, the system confirms the booking and assigns the selected slot to the user.

After reservation, real-time updates are pushed to all users through Firebase's real-time database synchronization. This ensures that the selected slot is immediately marked as occupied across all devices, maintaining data consistency and accuracy.

The system also incorporates user authentication using APSIT Moodle ID to restrict access to authorized users only. Notifications and alerts are sent using Firebase Cloud Messaging to inform users about booking confirmations, slot updates, or violations such as wrong parking.

Technically, the system is developed using Flutter for a responsive and user-friendly interface, while Firebase services (Authentication, Firestore, and Cloud Messaging) handle backend operations, real-time data updates, and communication. This architecture ensures scalability, reliability, and fast system performance.

Overall, the system integrates real-time querying, secure reservation processing, and continuous data synchronization to provide an efficient, organized, and user-friendly parking solution, reducing congestion and improving parking management within the APSIT campus.

2.1Features and Functionality

- Secure Login: Users log in using APSIT Moodle ID for authorized acces.
- Interactive Parking Map: Displays parking layout with real-time slot availability
- Slot Reservation: Allows users to reserve slots instantly based on vehicle type
- Parking Alerts: Sends notifications for wrong or restricted parking
- Zone Segregation: Differentiates between student and teacher parking areas

Chapter 3

Project Outcomes

The Smart Parking Management System significantly improves parking efficiency within the APSIT campus by providing a structured and digital approach to parking management. Users can securely log in using their APSIT Moodle ID and access real-time parking information, making it easier to locate and reserve available slots without unnecessary delays.

With the help of an interactive parking map, users can clearly view the layout of the parking area and identify free slots for cars and bikes. The system allows quick slot reservation, reducing the time spent searching for parking and minimizing traffic congestion inside the campus. It also supports vehicle-based selection and zone-wise allocation for students and staff, ensuring proper organization and fair usage of parking spaces.

Additionally, the system improves parking discipline by sending alerts and notifications for wrong parking or unauthorized usage. This helps in maintaining order and reducing misuse of parking areas. The implementation of real-time monitoring ensures better control and management of parking activities.

Overall, the project leads to optimized space utilization, reduced congestion, improved user convenience, enhanced security, and a more efficient and well-managed parking system within the APSIT campus.

Chapter 4

Software Requirements

Front-End:

- Flutter (Dart) for cross-platform mobile development
- Material Design 3 for UI
- Provider for state management
- Google Fonts (Poppins) for UI styling

Back-End:

- Cloud Firestore (NoSQL database)
- Firebase Authentication for login
- Firebase Realtime Database for real-time updates
- Firebase Cloud Messaging (FCM) for notifications

Platform Requirements:

- Android / iOS smartphone
- Internet connectivity

Chapter 5

Project Design

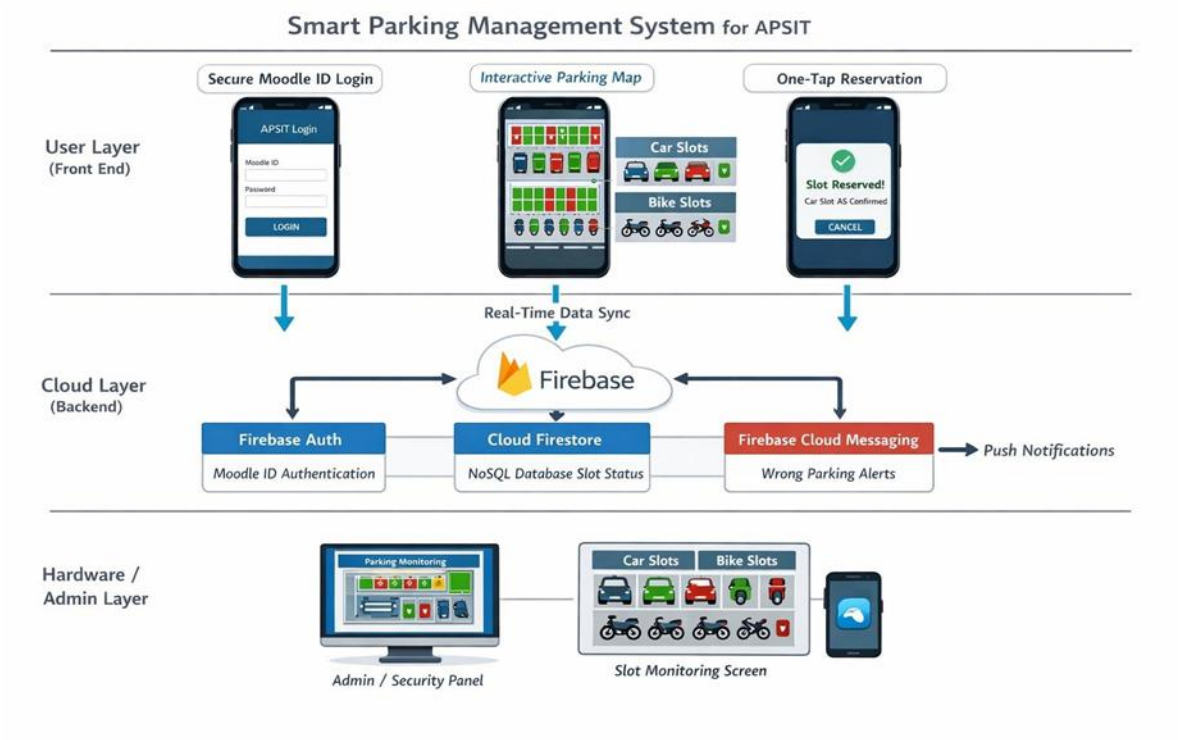


Fig 5 Workflow

Workflow Explanation

The Smart Parking Management System operates through a well-defined workflow that connects the user interface, cloud backend, and admin monitoring system to ensure smooth and efficient parking management.

- User Login and Authentication (User Layer)

The process begins when the user opens the mobile application and logs in using their APSIT Moodle ID. This login request is sent to Firebase Authentication, which verifies the user's credentials. Only authorized students and staff are allowed to access the system, ensuring security and controlled usage of parking facilities.

- Access to Interactive Parking Map

After successful login, the user is directed to the main dashboard where an interactive parking map is displayed. This map shows all available parking slots for cars and bikes, clearly marked and categorized. Different colors indicate the status of each slot (available, occupied, reserved), helping users quickly identify free spaces.

- Real-Time Data Synchronization

The parking slot data is continuously updated through Firebase Cloud Firestore and Realtime Database. Whenever a slot is booked or vacated, the change is instantly reflected across all users' devices. This real-time synchronization ensures that users always see accurate and up-to-date parking information, avoiding conflicts or double bookings.

- Slot Selection and Reservation

The user selects a suitable parking slot based on availability and vehicle type (car or bike). With a one-tap reservation feature, the selected slot is booked instantly. The system updates the database, marking the slot as occupied so that it is no longer visible as available to other users.

- Confirmation and Notifications

Once the booking is successful, the user receives a confirmation notification through Firebase Cloud Messaging (FCM). The system may also send alerts for important events such as reservation expiry, wrong parking, or violation of designated zones (student/teacher areas).

- Parking Monitoring and Management (Admin Layer)

On the backend, the admin or security panel continuously monitors parking activities. It displays real-time slot usage, identifies unauthorized parking, and ensures that users follow the rules. This layer helps maintain discipline and prevents misuse of parking spaces.

- Continuous Data Flow and System Efficiency

Throughout the entire process, data flows seamlessly between the user interface and the backend. Every action—login, reservation, cancellation, or update—is instantly processed and reflected in the system. This ensures fast response time, reliability, and efficient utilization of parking space.

Chapter 6

Project Scheduling

GANTT CHART TEMPLATE

Smartsheet T

A Gantt chart's visual timeline allows you to see details about each task as well as project dependencies.

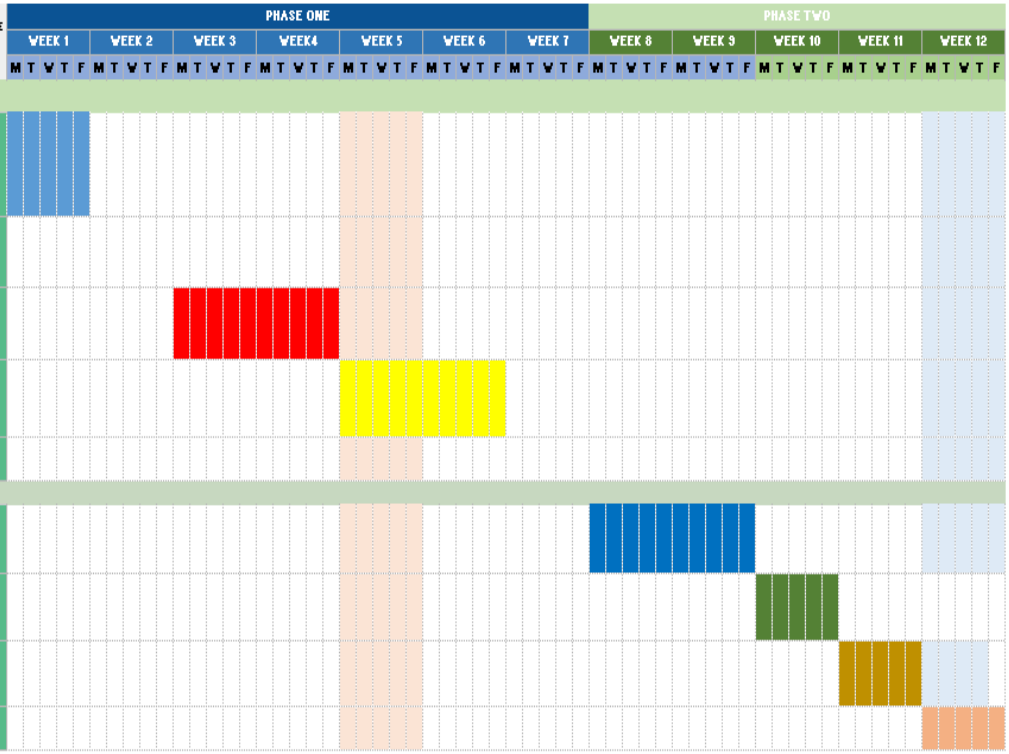
PROJECT TITLE: SMART PARKING MANAGEMENT SYSTEM FOR APSIT

PROJECT GUIDE: MS. HARSHA ZHOPE

INSTUTUTE & DEPARTMENT N: AP SHAH INSTITUTE OF TECHNOLOGY(CSE-Data Science)

DATE: 13/04/26

WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION(Weeks)	PERCENTAGE OF TASK COMPLETE
1	**Project Conception and Initiation**					
1.1	Group formation and Topic finalization. Identifying the scope and objectives of the Mini Project	All	1-13-26	1-16-26	1	100%
1.2	Identifying the functionalities of the Mini Project	All	1-16-26	1-20-26	1	100%
1.3	Discussing the project topic with the help of paper prototype.	All	1-20-26	2-3-26	2	100%
1.4	Designing the Graphical User Interface(GUI)	Yash Shinde and Shivansh Shukla	2-3-26	2-17-26	2	100%
1.5	Presentation I	All	2-17-26	2-24-26	1	100%
2	**Project Design and Implementation**					
2.1	Database Design	Yash Shinde and Swati Vishwakarma	2-24-26	3-10-26	2	100%
2.2	Database Connectivity of all modules	Shivansh Shukla and Shreyas Mhaske	3-10-26	3-17-26	1	100%
2.3	Integration of all modules and Report Writing	Swati Vishwakarma and Shreyas Mhaske	3-17-26	3-24-26	1	100%
2.4	Presentation II	All	3-24-26	4-10-26	1	100%



Chapter 7

Results

The implementation of the smart parking system resulted in:

- Faster parking slot allocation
- Reduced time spent searching for parking
- Improved parking discipline among users
- Efficient use of available parking space
- Real-time monitoring of parking activities
- Enhanced user experience through mobile application

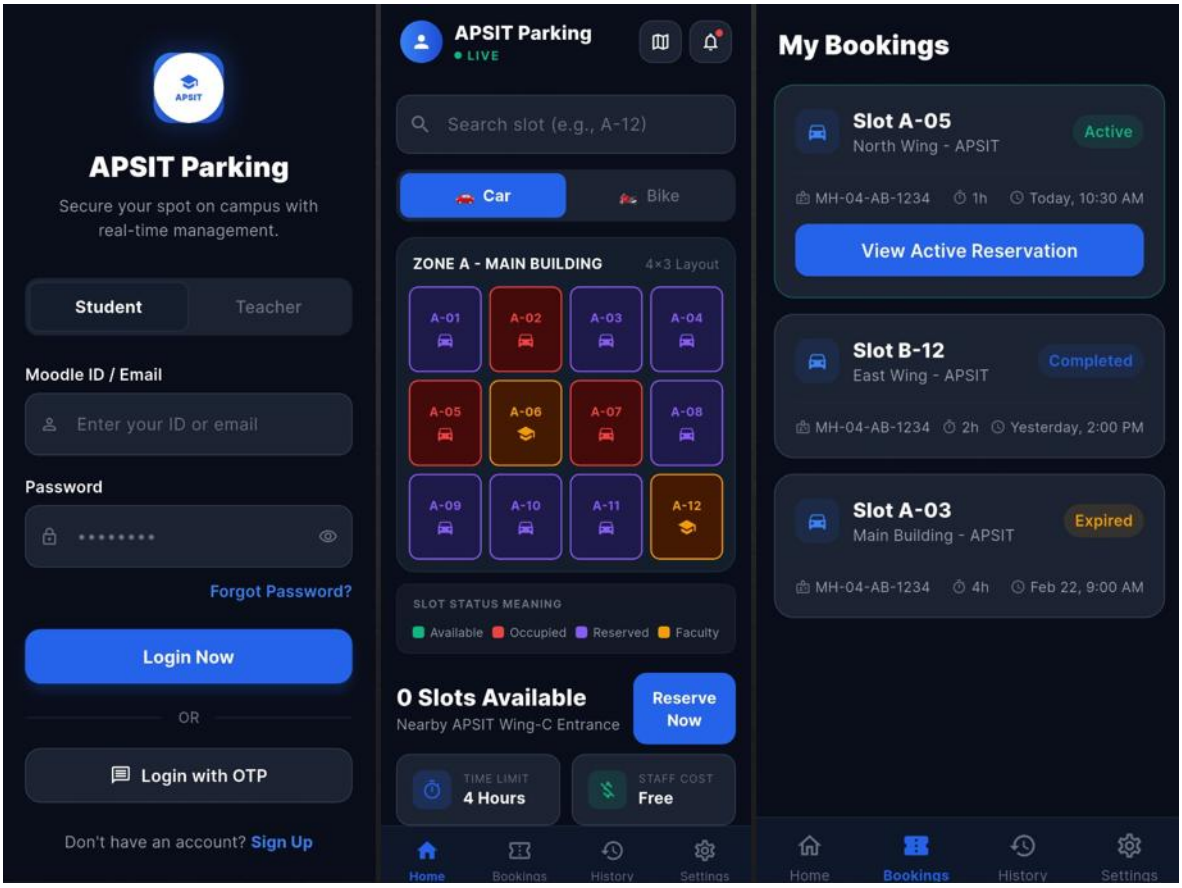


Fig 7.1

Figure 7.1 shows the main screens of the APSIT Smart Parking app: the profile/menu screen, parking slot selection screen, and booking/history screen. These screens allow users to navigate the app, view real-time slot availability, and manage their parking reservations efficiently.

a.)Login Screen

Allows users to securely log in using APSIT Moodle ID or email/password, ensuring authorized access.

b.)Parking Slot Selection Screen

Displays an interactive parking map with real-time slot availability. Users can easily select and reserve a slot.

c.)My Bookings Screen

Shows booking details like slot number, time, and status, allowing users to track and manage reservations.

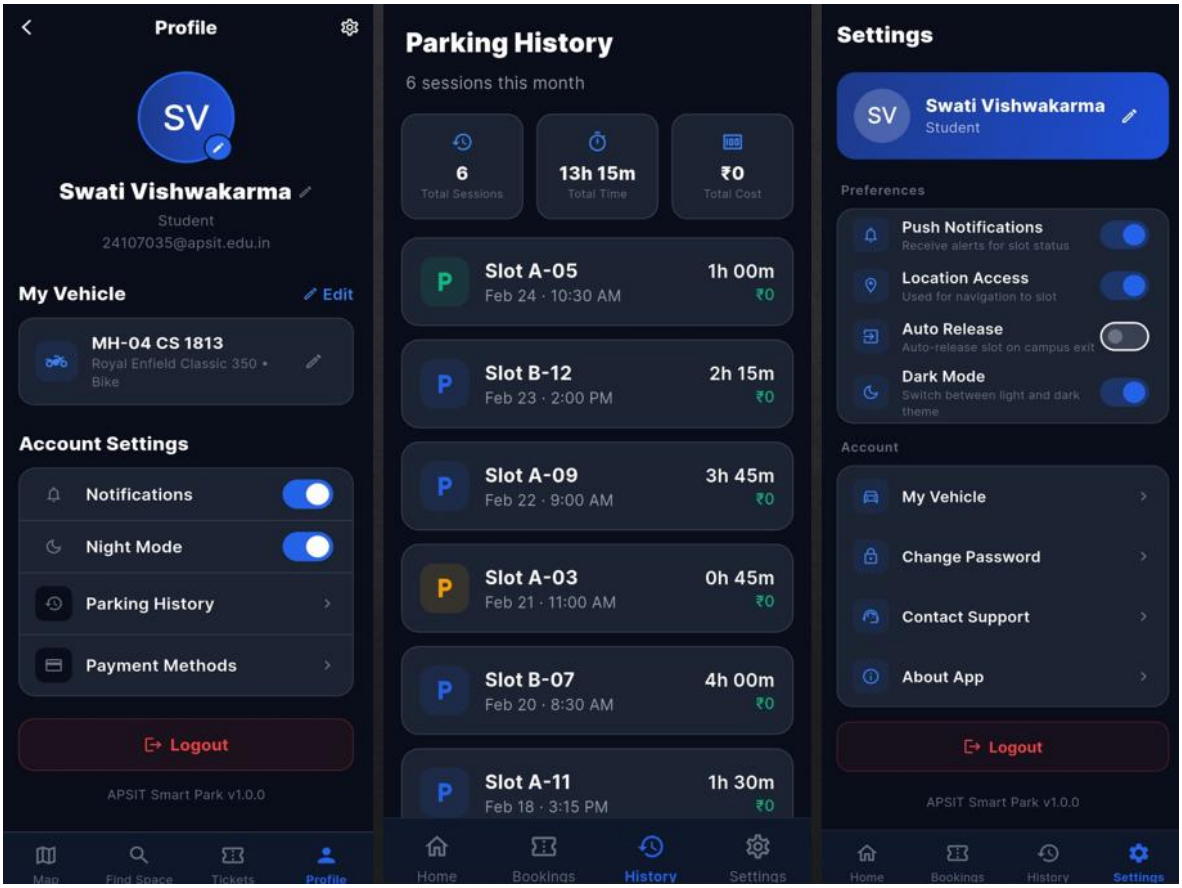


Fig 7.2

Figure 7.2 shows the profile screen, parking history screen, and settings screen of the APSIT Smart Parking app. These screens allow users to manage their account details, view past parking activity, and customize app preferences for a better user experience.

a.)Profile Screen

Displays user details and provides access to settings, parking history, and logout.

b.)Parking History Screen

Shows summary and list of past parking sessions with details like slot, time, and cost.

c.) Settings Screen

Allows users to manage preferences, account settings, and app options.

Chapter 8

Conclusion

The Smart Parking Management System provides an effective solution to the parking challenges faced within the APSIT campus. By using real-time data, digital mapping, and slot reservation features, the system simplifies the process of finding and managing parking spaces.

It helps in reducing congestion, saving time, and improving overall parking discipline among users. The use of modern technologies ensures accurate information, secure access, and smooth operation of the system.

In conclusion, this project demonstrates how a smart and automated approach can enhance campus management. With further improvements and scalability, the system can be expanded to larger environments and adapted for broader real-world applications.

Reference:

- Firebase Documentation
- Flutter Official Documentation
- Research papers on Smart Parking Systems
- APSIT campus parking observations